

16S Amplicon Analysis of Microbial Populations Using Qiime2

This workflow file provides step-by-step informatics methods. I've attempted to make the code easy to follow, but note that the output structure of some of these scripts, especially the Qiime2 scripts, doesn't match the results folder structure I returned in the SFTP folder; the scripts are designed for computational utility while the return folder is designed for human readability.

Cut Adapters and Trim/Filter by Quality

The goal of this script is to remove any adapter sequences from either end of the reads, which should decrease the number of reads that DADA2 flags as chimeric and removes from the downstream analysis. This is especially valuable for paired-end DADA2.

The files output to the cutadapt and unpaired fastq files are intermediates that were not returned or used downstream.

```
sbatch ./scripts-and-references/cut-trim-filter_args.sh \  
-i ./fastq \  
-c ./cut-fastq \  
-p ./trimmed-fastq \  
-u ./unpaired-fastq \  
-a ./scripts-and-references/all.fa -x 300
```

Check QC

After trimming off the adapters, I run QC checks on the trimmed and unprocessed fastq (to have a baseline for comparison). This gives valuable information about the remaining read counts, lengths, and quality, and verifies sufficient read depth before continuing the analysis.

```
sbatch ./scripts-and-references/fastqc_generation_args_phx.sh --fastqDir ./trimmed-fastq
```

Qiime2 Part 1: Denoising and Metrics

For this step, I imported the paired-end fastq reads into Qiime2 and carried out denoising using paired-end DADA2. If amplicon length had prevented DADA2 from merging the forward and reverse reads into longer amplicons for downstream analysis for a large percentage of reads, I would have used single-end DADA2 on just the forward reads; however, when possible paired-end DADA2 is more accurate and detects many more species due to the longer read length for matching against the reference database.

```
sbatch ./scripts-and-references/qiime2_pt1_args.sh \  
-f ./trimmed-fastq/fastq \  
-q ./qiime \  
-m ./scripts-and-references/metadata.txt \  
-p "p" -o "demux dada2 stats"
```

Qiime2 Part 2: Diversity and Taxonomy

Subsection 1

This script creates all the alpha and beta diversity metrics, as well as the taxonomic classifications and bar plots.

The value for `-x` is the smallest feature count for any of the samples in the set, obtained from the table QZV file created in the previous step. For this group I went with the second-lowest feature count because one sample had far fewer features than the

rest. However, the sampling cutoff is only applied for the alpha rarefaction curve analysis so omitting this sample won't affect the other analyses.

```
sbatch ./scripts-and-references/qiime2_pt2_args.sh \  
-q ./qiime \  
-r ./scripts-and-references/2024.09.greengenes.backbone.full-length.nb.sklearn-1.4.2.qza \  
-m ./metadata.txt \  
-c "Diarrhea_Presence Treatment" -n "Collection_Day Concentration 260_280_Ratio" \  
-s "p" -x 13701 -d 100
```

Subsection 2

Next, we performed ANCOM-BC analysis using the taxonomic classifications to determine differential abundances between groups in different categories. This is a more manual script so the project-specific variables are included in the code itself, at `qiime2_ancom_args.sh` in the scripts folder. Because of the pronounced batch effect caused by the Batch category (Batch 1 is clearly different from Batches 4-6), I used Batch as a covariate when modeling the statistical differences between the treatments.

ANCOM-BC results were calculated for all seven taxonomic levels, and both visual and tabular data are provided in the return folder.

Subsection 3

We also have included interactive sunburst plots to quickly visualize the taxonomic composition of each sample at different hierarchical levels. These were created using the included python script `sunburst_csvs_v2.py`.

```
python ./scripts-and-references/sunburst_csvs_v2.py \  
./qiime/rel-table6-paired-paired/ \  
rel-level6-table-paired-paired.tsv
```

As long as the `csvs` and `htmls` folders remain in the same directory and aren't renamed, the html files can be opened in your browser. Changing the names or separating the two folders will interfere with the json script that allows the plots to be viewable and interactive.

Additional Notes

Citations for Published Tools

FastQC

Andrews S, Krueger F, Segonds-Bichon A, Biggins L, Krueger C, Wingett S. FastQC: a quality control tool for high throughput sequence data. Babraham Institute. Published online 2010. <http://www.bioinformatics.babraham.ac.uk/projects/fastqc>

MultiQC

Ewels P, Magnusson M, Lundin S, Källér M. MultiQC: summarize analysis results for multiple tools and samples in a single report. Bioinformatics. 2016;32(19):3047-3048. doi:10.1093/bioinformatics/btw354

QIIME2

Bolyen E, Rideout JR, Dillon MR, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nat Biotechnol. 2019;37(8):852-857. doi:10.1038/s41587-019-0209-9

GreenGenes 16S rRNA Classifier

McDonald D, Jiang Y, Balaban M, et al. Greengenes2 unifies microbial data in a single reference tree. Nat Biotechnol. 2024;42(5):715-718. doi:10.1038/s41587-023-01845-1

Questions?

Please contact the core at asubioinformatics@asu.edu or kristina.buss@asu.edu if you have any questions or concerns about methods or results!